

Feature Comparison of Some Popular Boards

Board Name	Processor	Power	RAM	EEPROM	Storage	Peripherals	Extras
Raspberry Pi	ARM11 @ 700 MHz	2.5W-3.5W	256 MB	-	Expandable SD card	3.5mm jack, HDMI, USB 2.0, Ethernet RJ45	GPU handles 1080p video
Arduino Mega	ATMega1280 @ 16 MHz	0.2W	8 kB	4 kB	128 kB	54 digital I/O pins, 16 analogue inputs, 4 UARTS	USB, power jack
ArduinoDuemilanove	ATMega328 @ 16 MHz	0.2W	2 kB	1 kB	32 kB	14 digital I/O pins (six with PWM), 8 analogue input pins	Powered via USB, 7-12 volts
Arduino Mini	ATMega168 @ 16 MHz	0.2W	1 kB	512 bytes	16 kB	14 digital I/O pins, 8 analogue input pins	Breadboard mountable, pre-soldered pin headers
TI MSP430 Launchpad	MSP430 @ 25 MHz	0.28W	512 bytes	None	16 kB	10 GPIO	EEPROM emulation, 0.5m USB cable
STM32 Discovery	Cortex-M3 @ 72 MHz	NA	8 kB	4 kB	128 kB	Extension header for all QFP64 I/Os, STLink on-board	Two pushbuttons—user and reset, two user LEDs
chipKIT Uno32	PIC32 @ 80 MHz	NA	16 kB	None	128 kB	16-/32-bit timers, PWM, 16-channel 10-bit ADC, comparators, I ² C, SPI, UART	Same form-factor as Arduino Uno, compatible with most Arduino shields
BeagleBoard	ARM Cortex-A8 @ 600 MHz	2W	256 MB	-	256 MB	USB on-the-go, RS-232, JTAG, 3.5mm jacks, MMC slot with SDIO, DVI-D, S-Video	Alternative boot source button, boot from NAND, MMC, USB or serial
Atmel STK500	AVR	NA	NA	NA	2 MB	RS-232	Parallel and serial high-voltage mode, compatible with all Atmel chips
Freescale i.MX53 QSB	ARM Cortex-A8 @ 1 GHz	1W	1 GB	-	4GB MicroSD	7-pin SATA data connector, 10/100BT Ethernet, 2x high-speed USB, microUSB	MicroSD card slot, full-size card slot, JTAG connector, UART DB-9, 7.6x7.6cm ² (3x3-inch ²) 8-layer PCB

The chipKIT Uno32 development board costs just ₹ 2300 in India, compared to ₹ 2500 for the Arduino Uno on which it is based.

ATMEL STK500

This kit is shipped with an Atmel microcontroller in the socket marked SCKT3000D3. The default jumper settings allow the microcontroller to execute from the clock source and voltage regulator on the STK500 itself. However, the starter kit can be programmed for various clock and power sources.

The STK500 can be controlled from the AVR Studio (v3.2 or higher). AVR Studio is an integrated development environment for developing and debugging AVR applications. It also provides a project management tool, source file editor, simulator, in-circuit

emulator interface and programming interface for the board.

There also exists another model, the STK600. It has routing cards, but is far more expensive (at ₹ 17,330) than the STK500. The STK500 is available from element14 for around ₹ 7230. It was voted as the most popular developer board by the EFY Electronics Design Community on Facebook.

Freescale's i.MX53 Quick Start board

Freescale offers the Quick Start board targeting such applications as human-machine interface in embedded consumer, industrial and medical markets. The board comes integrated with the ARM Cortex-A8 processor and Freescale's own MC34708 PMIC. It also includes a display controller,

hardware-accelerated graphics, 1080p video decode, 720p video encode as well as some connectivity options. With Cortex-A8 clocked at 1 GHz and a full gigabyte of DDR3 memory, this board is built to perform.

The i.MX53 Quick Start board comes pre-installed with Linux OS. Android and Windows Embedded Compact 7 board support packages are also available through third parties. In addition to optimised board support packages, Freescale provides a large portfolio of optimised video, speech and audio codecs. Also included is a VMware player image running ready-to-go Linux. This allows those with a Windows PC to bypass the typical setup of a standard Linux-based development system. ●

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